

very low resistance, standby and shut-down current, and also protect against current-overload and short-circuit conditions in a circuit. On the other hand, the latest low-dropout regulators (LDOs) make less noise and offer high ripple rejection and low standby current.

"Moreover, current microprocessor supervisory circuits offer voltage-monitoring, power-up reset and watchdog functions. A combination of these solutions will enable engineers to build more efficient circuits that would consume much less power. For embedded designs, engineers can utilise microprocessor supervisory circuits, power switches and LDOs," explains Kay Annamalai, director of marketing, Pericom Semiconductor.

Another important aspect is the quiescent current of the components being used in the system. Quiescent current is the current flowing through the system or component when no load is being driven.

Abhishek Kumar, business development manager-power management, Texas Instruments (India), shares, "Quiescent current is important because battery life is determined by the total current drain composed of quiescent current (mainly in standby conditions) and load current. Quiescent current consumption should be as low as possible in order to prolong the battery's life."

Improve your power system performance. Faster, lower-loss switching topologies such as zero current switching, zero voltage switching and sine-amplitude conversion are allowing significant improvements in power system performance.

"Reductions in the overall size of power components and increases in conversion efficiency are opening up opportunities for integrating power management solutions adjacent to their demanding loads, microcontrollers and FPGAs, right on the main PCB," informs Andy Gales, vice-president, international sales, Vicor Corporation.

Power system designers are demanding even higher input voltage

Tuning the code

Mobile devices typically have their operating systems tuned for better battery life by making several special features available to designers. Android utilises a system called 'wakelocks,' which is a set of patches to the Linux kernel that allow a caller to prevent the system from going to low-power state.

"If a novice programmer created a buggy code where it would acquire wakelock and then get into some loop somewhere, the device would hold wakelock and never go into standby mode," explains Darshak Vasavada, CEO, Stamp Computers.

It is a myth that only hardware dissipates power, and software does not play any role. In all the embedded systems it is the software that controls the hardware and thus plays an important part in optimising the power of the system as a whole. Nitin Gupta, lead engineer, μ Energy Applications, CSR India explains where and how to optimise. "Optimise the memory accesses in the software, as more frequent are the accesses, more time is used up in switching high capacity bus system and also the memory system. You can also use parallel processing of the memory system to access more data in one access. Another tip is that all the IPs should be maintained in no-clock/lower-clock, if they are not required in the application so that the power dissipation is least, but it's a trade-off with performance."

ranges from their component power suppliers. Vicor recently introduced a wide-input-range zero voltage switching buck regulator that accepts inputs up to 36V, yet provides a 1.8V, 15A output at efficiencies typically exceeding 90 per cent—all in a 14x10x2.6mm³ SIP. The technology to achieve this performance was developed in-house and utilises a clamp-switch to minimise losses in switching the main FET.

Today, battery-system designs typically have at least one of the following integrated circuits (ICs), in addition to the system's main chipset (if applicable):

1. Power management unit
2. Microcontroller unit
3. Battery management unit

The power management unit offers regulated voltage or current in a system.

"Some voltage regulations are also integrated into the main chipset. However, a monolithic converter is still required due to the complexity of the layout, EMI concerns and lack of performance, including an insufficient amount of power-output channels or load capabilities, etc. Monolithic voltage converters are available with single output and multiple outputs," adds Brian.

To know how to select low-power MCUs for your project, you may please refer to "How to Select A Microcontroller" article published in Dec. 2012 issue of EFY.

Computing elements and scaling clock rates. Power consumption by processor cores can be controlled in a big way by utilising the various sleep modes available. Advanced Micro Devices' Larne reference platform is projected to measure APU power at 1.2W during idle mode, 1.4W during Web browsing, 2.35W during video playback and 0.02W during a system's sleep state.

Abhishek advises design engineers to keep in mind many factors in their system architecture and design during development.

"More specifically, engineers must take well-informed system architecture decisions, choose optimal components and provide microcontrollers that optimise functionality, resulting in a longer battery life. This is definitely much more complex than simply estimating how much current the CPU consumes when active. Depending on the application being developed, standby current, peripheral current or data logging current may have a more significant impact on battery life than CPU power," he shares.

The second aspect that increases the battery life is dynamic voltage scaling. "It is a framework to change the frequency and/or operating voltage of a processor based on system performance requirements at a given point of time. Power devices with dynamic voltage scaling as an inbuilt feature will help lower power consumption